Vandermonde's Identity

- **Theorem (Vandermonde's identity)**

Let m, n, and r be integers with $0 \leq r \leq m$ and $0 \leq r \leq n$. Then

$$\binom{m+n}{r} = \sum_{k=0}^{r} \binom{m}{r-k} \binom{n}{k}.$$

- Discovered by Zhu Shijie in 1303.